Epitranscriptomic Determinants of RNA Binding Protein Specificity: A Comprehensive Computational and Experimental Framework to Bridge In Vitro and In Vivo Binding Landscapes

1. Introduction: The Specificity Paradox in Post-Transcriptional Regulation

The precise regulation of gene expression extends far beyond transcriptional control; it is meticulously orchestrated at the post-transcriptional level by a vast array of RNA Binding Proteins (RBPs). These proteins serve as the custodians of the transcriptome, governing every phase of an RNA molecule's life cycle, including pre-mRNA splicing, 3' end processing, nuclear export, subcellular localization, translational efficiency, and eventual decay. The human genome encodes approximately 1,500 RBPs, each possessing distinct RNA Binding Domains (RBDs) such as the RNA Recognition Motif (RRM), the K-Homology (KH) domain, and the Zinc Finger (ZnF) domain. The central dogma of RBP biology posits that the functional output of an RBP is dictated by its binding specificity—the set of thermodynamic rules that determine which RNA sequences and structures it preferentially engages.

Over the past decade, two methodological paradigms have emerged to decode these rules: high-throughput *in vitro* affinity assays and transcriptome-wide *in vivo* occupancy profiling. The study by Dominguez et al. (2018), titled *Sequence, Structure, and Context Preferences of Human RNA Binding Proteins*, represents a landmark achievement in the former category.[1] By applying RNA Bind-n-Seq (RBNS) to 78 human RBPs, the authors generated comprehensive "affinity landscapes" based on the binding of recombinant RBDs to randomized synthetic RNA libraries. This dataset provides a pristine view of intrinsic specificity, unencumbered by the complexities of the cellular environment. Conversely, the Enhanced Crosslinking and Immunoprecipitation (eCLIP) method, championed by the Yeo lab and the ENCODE consortium, provides a snapshot of RBP binding *in situ*, capturing interactions as they occur within the crowded, competitive, and compartmentalized milieu of the living cell.[2]

1.1 The Divergence Between Thermodynamic Potential and Cellular Reality

A persistent and confounding observation in RNA biology is the discrepancy between the motifs identified by these two approaches. While RBNS and eCLIP data sets often show statistically significant overlap, the correlation is rarely perfect. High-affinity motifs identified *in vitro* frequently remain unoccupied *in vivo* ("missing peaks"), while RBPs often bind sequences in cells that appear to be low-affinity or non-specific targets in purely thermodynamic assays ("unexplained binding"). The analysis by Dominguez et al. (2018) highlighted this divergence for a subset of 24 RBPs profiled by both methods, noting that while intrinsic affinity is a major determinant, it is insufficient to fully predict cellular occupancy.[1]

Standard models attribute these discrepancies to three primary factors:

1. **RNA Secondary Structure:** An RNA motif may be thermodynamically perfect for binding but sequestered within a stable double-stranded helix or G-quadruplex *in vivo*, rendering it inaccessible to single-strand binding proteins.
2. **Competitive and Cooperative Binding:** The cellular environment is replete with competing RBPs battling for overlapping sites, as well as co-factors that may be required to stabilize weak interactions (cooperativity) that are absent in purified RBNS assays.
3. **Subcellular Localization:** An RBP sequestered in the nucleus cannot bind a high-affinity motif present only on a cytosolic transcript.

However, your hypothesis introduces a fourth, transformative variable: the **Epitranscriptome**. Cellular RNA is not a simple polymer of Adenine, Cytosine, Guanine, and Uracil. It is a chemically diverse landscape decorated with over 170 distinct post-transcriptional modifications, ranging from the abundant N6-methyladenosine (m6A) to the structure-altering Pseudouridine (Ψ) and the stabilizing 5-methylcytosine (m5C).

Consultation Point 1: *Theoretical Foundation*

We are proceeding with the core hypothesis that the use of synthetic, unmodified RNA in RBNS represents a "blind spot" in the assay. By comparing unmodified RBNS data with endogenous eCLIP data, we effectively have a differential experiment: (Sequence + Structure + Modifications) - (Sequence + Structure) = Modification Effect.

Does this formulation align with your intent? specifically, do you view RNA modifications as primarily binary switches (on/off) or as rheostats that fine-tune affinity? The computational model design will differ based on whether we search for "presence/absence" or "quantitative shifts" in binding energy.

1.2 The Epitranscriptome as the Missing Specificity Determinant

The chemical moieties added to RNA bases can fundamentally alter the physicochemical

interface presented to an RBP.

- **N6-methyladenosine (m6A):** This modification involves the addition of a methyl group to the N6 position of adenosine. It is the most prevalent internal mRNA modification and is typically deposited at DRACH consensus motifs ($D = A/G/U$, $R = A/G$, $H = A/C/U$).[5] The methyl group can sterically clash with RBPs that require Watson-Crick signaling or, conversely, create a hydrophobic handle for "reader" proteins containing YTH domains. Furthermore, m6A destabilizes RNA duplexes (the "m6A-switch"), potentially exposing motifs that would otherwise be buried in secondary structures.
- **Pseudouridine (Ψ):** Often termed the "fifth nucleotide," Ψ is an isomer of uridine coordinated by pseudouridine synthases (PUS enzymes) such as PUS7 (recognizing UGUAR) and TRUB1 (recognizing GUUCN).[6] The C-C glycosidic bond in Ψ provides an extra hydrogen bond donor, enhancing base stacking and rigidifying the sugar-phosphate backbone. This could lock RNA into conformations that either favor or disfavor RBP docking.
- **5-methylcytosine (m5C):** Enriched in GC-rich regions and catalyzed by NSUN enzymes (e.g., NSUN2), m5C can enhance base stacking and thermal stability.[8] Like m6A, it creates a unique chemical signature that can be read by specific proteins (e.g., ALYREF) while repelling others.
- **N4-acetylcytidine (ac4C):** A recently characterized modification on mRNA, deposited by NAT10 at CCG motifs, which enhances translation and stability.[10]

Since the RBNS assay described in Dominguez et al. utilized *in vitro* transcription (IVT) with standard nucleotides, the resulting "affinity landscape" is blind to these chemical interactions. If an RBP is an "m6A-reader," RBNS will identify the underlying sequence as a weak binder (or non-binder), whereas eCLIP will show robust occupancy. This specific type of divergence—high cellular occupancy vs. low intrinsic affinity—constitutes a "Modification Signature" that we can computationally hunt for.

1.3 Scope of the Report

This document serves as a comprehensive architectural blueprint for your research. It is designed to be fed into "Claude Code" to generate the executable bioinformatics pipelines and experimental protocols. The report is divided into:

1. **Data Selection:** Rigorous identification of the 24 "Gold Standard" RBPs and the requisite datasets.
2. **Specific Aim 1 (Computational):** A three-stage pipeline to identify "Discrepant Peaks," intersect them with epitranscriptomic maps, and model the structural basis of the interaction using AlphaFold3.

3. **Specific Aim 2 (Experimental):** A detailed design for a "Plasmid-based Endogenous RBNS" assay to validate these findings *in vitro* using cell-derived RNA.

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2. Dataset Definition: The "Gold Standard" Overlap

To operationalize this study, we must first curate the precise list of proteins and datasets. The Dominguez et al. (2018) study focuses on 78 RBPs, but only a subset overlaps with the high-quality "reproducible" eCLIP datasets generated by the Yeo Lab for the ENCODE consortium.

2.1 The 24 Overlap RBPs

Based on a cross-reference of the RBNS dataset [1] and ENCODE eCLIP metadata [4], the following 24 RBPs possess paired data suitable for this analysis. We have categorized them by domain architecture, as this influences their potential mode of modification recognition.

Table 1: The "Gold Standard" RBP Cohort

RBP Name	Domain Type	RBNS Accession (Aggregated)	eCLIP Accession (K562/HepG2)	Primary RBNS Motif (In Vitro)	Known Biology/Notes
HNRNPC	RRM	ENCSR876DCD	ENCSR456FVU	Poly-U (U_5)	Comparison with TIA1 (also Poly-U) highlights context dependence.[1]
TIA1	RRM	ENCSR876DCD	ENCSR456FVU	Poly-U (U_5)	Stress granule component; binds U-rich sequences.

HNRNPK	KH	ENCSR876DCD	ENCSR456FVU	Poly-C (C_6)	Major axis of C-rich binding; known to be sensitive to structure.
PCBP2	KH	ENCSR876DCD	ENCSR456FVU	Poly-C (C_6)	Paralog of HNRNPK; often binds C-rich tracts.
RBFOX2	RRM	ENCSR876DCD	ENCSR456FVU	GCAUG	Highly specific splicing factor; motif is "gold standard" for comparisons.
PTBP3	RRM	ENCSR876DCD	ENCSR456FVU	C/U-rich	Paralog of PTBP1; regulates splicing and stability.
TARDBP	RRM	ENCSR876DCD	ENCSR456FVU	UG repeats	TDP-43; ALS-associated; binds long UG repeats.

IGF2BP1	KH	ENCSR876DCD	ENCSR456FVU	CAU / CA rich	**Known m6A Reader** (anti-reader/stabilizer). Positive control.
IGF2BP2	KH	ENCSR876DCD	ENCSR456FVU	CAU / CA rich	**Known m6A Reader**. Positive control for mod-dependency.
QKI	KH	ENCSR876DCD	ENCSR456FVU	ACUAAY	Quaking homolog; binds specific bipartite motifs.
SRSF1	RRM	ENCSR876DCD	ENCSR456FVU	GGA / Purine-rich	Splicing factor; motif often creates secondary structure.
SRSF9	RRM	ENCSR876DCD	ENCSR456FVU	GGA / Purine-rich	SR protein; likely sensitive to purine modificatio

					ns (m6A/m1A).
RBM22	RRM	ENCSR876DCD	ENCSR456FVU	Stem-Loop / G rich	Binds structured RNA in spliceosome; structure might be mod-regulated.
TRA2A	RRM	ENCSR876DCD	ENCSR456FVU	GAA repeats	Splicing regulator.
HNRNPL	RRM	ENCSR876DCD	ENCSR456FVU	AC repeats	Binds CA-repeat microsatellites.
LIN28B	CSD/ZnF	ENCSR876DCD	ENCSR456FVU	GGAG / Let-7 loop	Regulates Let-7 processing; highly structure-dependent.
ZNF326	ZnF	ENCSR876DCD	ENCSR456FVU	Structured A/G	Known to prefer stem-loops.[1]
FUS	RRM/ZnF	ENCSR876DCD	ENCSR456FVU	GGUG / Loop	ALS-associated; prone to aggregation/phase

					separation.
MATR3	RRM	ENCSR876DCD	ENCSR456FVU	Pyrimidine-rich	Nuclear matrix protein.
HNRNPA1	RRM	ENCSR876DCD	ENCSR456FVU	UAGGG	Splicing repressor.
HNRNPM	RRM	ENCSR876DCD	ENCSR456FVU	G / U rich	Splicing factor.
NONO	RRM	ENCSR876DCD	ENCSR456FVU	AG rich	Paraspeckle component.
U2AF2	RRM	ENCSR876DCD	ENCSR456FVU	Poly-U / Py-tract	Binds 3' splice site polypyrimidine tract.
EWSR1	RRM/ZnF	ENCSR876DCD	ENCSR456FVU	G rich	FET family protein; similar to FUS/TAF15.

Consultation Point 2: *Data Selection Strategy*

The list includes **IGF2BP1/2**, which are validated m6A readers. This is advantageous as they serve as "Positive Controls"—our pipeline *must* identify them as having modification-dependent binding.

Are you satisfied with focusing on this list of 24, or would you like to expand to the full 78 RBNS proteins by inferring eCLIP data from other sources (e.g., POSTAR3)? Sticking to the ENCODE gold-standard set (24 RBPs) is recommended for statistical rigor, but expanding could increase discovery potential.

2.2 Epitranscriptomic Map Integration

To correlate discrepancies with modifications, we need high-confidence maps of modification sites in the same cell lines used for eCLIP (K562 and HepG2). Mixing cell lines (e.g., using HEK293 modification maps with K562 binding data) is a major confounder, as RNA modifications are highly dynamic and cell-type specific.

Database Strategy:

1. **REPIC (RNA Epitranscriptome Collection):** This database is critical for m6A analysis. It aggregates ~50 studies and provides "consensus" m6A peaks filtered by cell line.[13]
   - *Action:* Download K562 and HepG2 specific m6A BED files.
2. **RMBase v2.0:** A comprehensive resource for non-m6A modifications (Ψ, m5C, m1A, 2'-O-Me).[15]
   - *Action:* Download cell-line specific tracks. Apply a filter to retain only sites supported by at least two independent studies or methods (e.g., Pseudo-seq AND Ψ-seq).
3. **Motif-Guided Filtering:** To remove false positives from low-resolution immunoprecipitation data (like MeRIP-seq), we will intersection modification peaks with their respective consensus motifs:
   - **m6A:** Filter for DRACH ($D = A/G/U$, $R = A/G$, $H = A/C/U$).[5]
   - **Ψ:** Filter for UGUAR (PUS7) and GUUCN (TRUB1).[7]
   - **m5C:** Filter for NSUN2 consensus (variable, often C-rich or NGG).[17]
   - **ac4C:** Filter for CCG.[10]

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3. Specific Aim 1: Computational Modeling Pipeline

This aim focuses on identifying and characterizing the "Specificity Gap" using data science and structural modeling. We will deploy a modular pipeline, which you can implement using Claude Code.

3.1 Module A: The "Discrepancy Hunter" Algorithm

The core of this aim is to mathematically define when an RBP "should" bind but doesn't (or vice versa).

Methodological Workflow:

1. **Data Harmonization:**
   - **RBNS Data:** Convert the 5-mer/6-mer Z-scores from Dominguez et al. into a dictionary RBNS_dict = {kmer: z_score}. Normalize scores to a 0-1 scale to allow comparison across proteins.
   - **eCLIP Data:** Import the ENCODE "reproducible" narrowPeak files. These files

represent high-confidence binding sites (IDR < 0.05).
    - **Sequence Extraction:** Extract the genomic sequence for each eCLIP peak, extending it to a fixed window (e.g., ± 50 nt from the peak summit) to capture the relevant motif context.
2. **The Scoring Function (RBNS-to-eCLIP Mapping):**
   We need to ask: *Does the sequence under this eCLIP peak contain a high-affinity RBNS motif?*
    - For each eCLIP peak sequence S, sliding a window of size k (e.g., 5 or 6) across the sequence.
    - Calculate the **Maximal Affinity Score ($Score_{max}$)**:

$$Score_{max} = \max_i (Z_{RBNS}(S[i:i+k]))$$

    - Calculate the **Cumulative Affinity Score ($Score_{sum}$)**:

$$Score_{sum} = \sum_i (Z_{RBNS}(S[i:i+k]))$$

    - *Rationale:* Some RBPs bind a single high-affinity site ($Score_{max}$ is better), while others rely on avidity sites with multiple weak motifs ($Score_{sum}$ is better). We will compute both.
3. **Defining the Populations:**
   Based on the distribution of $Score_{max}$ across all eCLIP peaks, we will classify peaks into three bins:
    - **Bin 1: Canonical Binding (Explained).** High eCLIP enrichment AND High RBNS Score (Top quartile). These are standard binding events.
    - **Bin 2: The "Gain-of-Function" Candidates (Discrepant).** High eCLIP enrichment BUT Low RBNS Score (Bottom quartile).
        - *Hypothesis:* These sequences appear "weak" to the RBP in unmodified form (RBNS), but the RBP binds strongly *in vivo*. **We hypothesize these are enabled by modifications (e.g., m6A recruitment).**
    - **Bin 3: The "Loss-of-Function" Candidates.** (Requires generating a "negative" set of genomic regions that have High RBNS scores but *no* eCLIP peaks).
        - *Hypothesis:* These high-affinity motifs are masked *in vivo* by modifications (e.g., steric hindrance).

Consultation Point 3: *Scoring Metrics*

We plan to use a sliding window approach ($k = 5$ or 6). Dominguez et al. showed that flanking sequences also matter.

Should we include a "Flanking Context" weight in our score? For example, if a motif is flanked by U-rich sequences, should it get a bonus? This increases complexity but might improve the "Explained" baseline, isolating true modification candidates more effectively.

3.2 Module B: Epitranscriptomic Intersection and Enrichment

Once Bin 2 (Gain-of-Function) is identified, we test if it is enriched for modifications.

Statistical Approach:

1. **Intersection:** Use bedtools intersect to map the K562/HepG2 modification sites (m6A, m5C, Ψ) to the eCLIP peaks in Bin 1 (Control) and Bin 2 (Test).
2. **Enrichment Calculation:** Calculate the Odds Ratio (OR) for each modification type:

$$OR = \frac{(\text{Peaks}_{\text{Bin2, Mod+}} / \text{Peaks}_{\text{Bin2, Mod-}})}{(\text{Peaks}_{\text{Bin1, Mod+}} / \text{Peaks}_{\text{Bin1, Mod-}})}$$

3. **Significance Testing:** Use Fisher's Exact Test to determine if the enrichment is significant ($p < 0.05$, Bonferroni corrected).

The "Hidden Motif" Discovery:

For peaks in Bin 2 that *do* intersect with a modification, we will extract the sequence underlying the modification site (e.g., the DRACH motif) and align them.

- *Outcome:* If we align the "unexplained" binding sites of RBP-X and find they all center on a GGACU (m6A) consensus rather than the RBP's canonical motif, this is strong evidence of m6A-dependent binding.

3.3 Module C: Structural Modeling with AlphaFold3

You expressed a strong interest in using AlphaFold3 (AF3) to model these interactions, specifically utilizing its ability to handle covalent modifications. This is the mechanistic validation step.

Target Selection:

We will select the top 5 "Discrepant Motifs" identified in Module B. For example, if RBP-X binds UGUAR sites *in vivo* (but not *in vitro*) and these sites are Ψ-modified.

AF3 Workflow for Claude Implementation:

1. **Input Preparation:** AF3 requires JSON input specifying the polymer chains and modifications.
   - **Protein:** Full sequence of the RBP (or the RBD used in RBNS).
   - **RNA (Control):** The unmodified RNA sequence (e.g., ...UGUAR...).
   - **RNA (Test):** The modified RNA sequence. We must utilize the Chemical Component Dictionary (CCD) codes to specify the modification.
     - m6A Code: 6MZ
     - Pseudouridine Code: PSU or P
     - m5C Code: 5MC
2. **Modeling Strategy:**
   - Dock the RNA to the RBP using AF3 in two parallel runs: Unmodified vs. Modified.
   - *Constraint:* If structural data exists (e.g., PDBs of the RBP), use them as templates to guide the fold.
3. **Analysis of Output:**
   - **PAE (Predicted Aligned Error):** Compare the PAE matrix at the protein-RNA interface. A lower PAE (higher confidence) in the Modified structure vs. the Unmodified structure suggests the modification stabilizes the complex.
   - **Interface Energy (ΔG):** Use a post-processing tool (like FoldX or Rosetta) on the AF3-predicted PDB structures to calculate the theoretical binding energy.
   - **Hydrogen Bonding Analysis:** specifically check if the modification (e.g., the N6-methyl group or the N1-H of Ψ) forms new H-bonds or van der Waals contacts with amino acid side chains.

Consultation Point 4: *Structural Complexity*

AlphaFold3 is state-of-the-art but computationally expensive and "hallucination-prone" for novel ligands.

Would you like to include a Molecular Dynamics (MD) equilibration step? We could take the static AF3 structure and run a short (100ns) simulation to see if the modified RNA stays bound while the unmodified RNA drifts away. This adds computational cost but significantly strengthens the physical argument.

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4. Specific Aim 2: Experimental Design – The "Endogenous RBNS"

To empirically validate the computational predictions, we must move beyond synthetic RNA. You proposed encoding the library on plasmids to allow cellular processing. This is a brilliant strategy, often called "Cellular RBNS" or "In Vivo RBNS," but it presents unique engineering

challenges regarding 3' end heterogeneity and library complexity.

4.1 The Plasmid Library Architecture

Design Philosophy:

Standard RBNS uses random N_{20} oligomers. If we express N_{20} in a plasmid, the complexity ($4^{20} \approx 10^{12}$) exceeds the transformation efficiency of mammalian cells ($\approx 10^7 - 10^8$). We cannot screen *all* sequences. Therefore, we must design a **Biased Library**.

The Construct:

1. **Promoter: CMV** (Cytomegalovirus) promoter for high-level expression by RNA Polymerase II.
   - *Why Pol II?* Pol II transcripts undergo capping, splicing, and are heavily trafficked through the nuclear machinery where most modification writers (METTL3, NSUN2) reside. T7 transcripts (cytoplasmic) would miss these marks.
2. **The "Bait" Insert (150nt total):**
   - **5' Adapter:** Fixed sequence for PCR/Sequencing.
   - **Variable Region (40nt):** This will be a synthesized pool containing:
     - 2,000 "Canonical" RBNS motifs (Positive Control).
     - 2,000 "Discrepant" motifs identified in Aim 1 (The Test Set).
     - 2,000 Negative Control sequences (genomic regions with no binding).
     - 2,000 Mutated Discrepant motifs (e.g., DRACH $\rightarrow$ DRGCH to abolish methylation).
   - **3' Adapter:** Fixed sequence for PCR/Sequencing.
3. **Structural Control Element: The Ribozyme:**
   - *The Problem:* Pol II termination and polyadenylation are imprecise. They add variable-length Poly-A tails that would confuse the motif calling.
   - *The Solution:* Encode a **Hepatitis Delta Virus (HDV) Ribozyme** or **Hammerhead Ribozyme** immediately downstream of the 3' adapter.
   - *Mechanism:* Upon transcription, the ribozyme self-cleaves at a precise phosphodiester bond, generating a distinct 3' end for every transcript in the library, independent of the variable poly-A tail machinery. This mimics the sharp 3' end of synthetic RBNS oligos.

Consultation Point 5: *Library Design*

We propose using a Biased Library (Chip-based synthesis, e.g., Twist Bioscience or Agilent) rather than a degenerate N20 oligo. This allows us to specifically test the "Discrepant" candidates found in Aim 1. Do you agree with this targeted approach, or do you prefer a

degenerate library to discover completely novel modification-dependent motifs, accepting that we will undersample the sequence space?

4.2 The "Endogenous RBNS" Workflow

Step 1: Transfection and Modification

- Transfect the plasmid library into **HEK293 cells** (wild-type).
- *Biological Replicate:* Transfect into **Modification-Depleted Cells** (e.g., METTL3-Knockdown or cells treated with 3-Deazaadenosine). This is the crucial control you asked for. If binding is modification-dependent, it should disappear in this cell line.

Step 2: RNA Harvesting

- Wait 24-48 hours for transcription and modification.
- Lyse cells and perform **Total RNA Extraction**.
- **Size Selection:** Run on a denaturing PAGE gel and excise the band corresponding to the ribozyme-cleaved library insert (~100nt). This removes endogenous mRNA and rRNA background.

Step 3: The RBNS Assay

- Incubate this *cell-derived, modified* RNA pool with the recombinant, SBP-tagged RBP (just like standard RBNS).
- **Pull-down:** Streptavidin beads.
- **Library Prep:** RT-PCR using the 5'/3' adapters.
- **Sequencing:** Illumina NextSeq.

Step 4: Analysis

- Calculate Enrichment (R) for each sequence in the design.
- **The "Delta-R" Metric:** Compare R values between conditions.
  - $\Delta R_{\text{mod}} = R_{\text{WT Cells}} - R_{\text{Synthetic RBNS}}$
  - $\Delta R_{\text{writer}} = R_{\text{WT Cells}} - R_{\text{METTL3-KD Cells}}$
- *Conclusion:* A sequence with high ΔR_{mod} and high ΔR_{writer} is confirmed as an m6A-dependent binder.

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5. Detailed Implementation Plan for Claude Code

This section provides the specific "Prompts" and logic structures you should use when interacting with Claude Code to generate the actual scripts and protocols.

5.1 Prompting for Data Pipeline (Aim 1)

User Action: Copy/Paste the following structural instructions to Claude.

> "I need a Python pipeline to analyze RBP binding discrepancies.
>
> **Inputs:**
>
> 1. A table of RBNS Z-scores for 5-mers (from Dominguez 2018).
> 2. BED files of eCLIP narrowPeaks (from ENCODE) for K562 cells.
> 3. BED files of m6A sites (from REPIC) and Pseudouridine sites (from RMBase).
>
> **Task 1: Scoring.** Write a function calculate_motif_score(sequence, z_score_dict) that slides a 5-nt window across a sequence and returns the maximum Z-score found.
>
> **Task 2: Classification.** Write a script using pybedtools to:
>
> - Extract sequences for all eCLIP peaks.
> - Score them using the function above.
> - Plot a histogram of scores.
> - Filter for peaks with Score < 2.0 (Low Affinity) but Signal_Value > 10 (High In Vivo Binding). Save these as discrepant_peaks.bed.
>
> **Task 3: Enrichment.** Perform a Fisher's Exact Test to see if discrepant_peaks.bed overlaps with m6A sites significantly more often than canonical_peaks.bed (Score > 2.0).
>
> Use pandas, scipy.stats, and pybedtools."

5.2 Prompting for AlphaFold3 Setup (Aim 1)

User Action: Generate the JSONs for structural modeling.

> "I need a Python script to generate input JSON files for AlphaFold3.
>
> **Inputs:**
>
> 1. Protein Sequence:
> 2. RNA Sequence: 'UGUAR' motif embedded in a 15nt hairpin.
>
> **Requirements:**
>
> - Generate two JSON files: unmodified.json and modified.json.
> - In modified.json, replace the central Adenosine residue in the RNA chain with the chemical component code 6MZ (N6-methyladenosine).
> - Ensure the format complies with the AlphaFold Server input schema."

5.3 Prompting for Experimental Protocol (Aim 2)

User Action: Generate the specific sequences for the plasmid library.

> "I need to design an oligonucleotide library for a plasmid-based RBNS assay.
>
> **Components:**
>
> 1. 5' Adapter: GGGGAGUUCUACAGUCCGACGAUC (Illumina RP1)
> 2. 3' Adapter: UGGAAUUCUCGGGUGUCAAGG (Illumina RP2)
> 3. HDV Ribozyme Sequence: GGCCGGCAUGGUCCCAGCCUCCUCGCUGGCGCCGGCUGGGCAACAUCCUGAGGGGACCGUCCCC
>
> **Task:**
>
> Write a Python script to generate a CSV file of 10,000 oligo sequences.
>
> - The script should take a list of 5-mer motifs (e.g., ['UGUAA', 'GGACU']).
> - For each motif, generate 100 variations with random flanking bases (N) such that the total length of the variable region is 40nt.
> - Concatenate: Promoter_Sequence + 5_Adapter + Variable_Region + 3_Adapter + HDV_Ribozyme.
> - Check that no internal restriction sites (EcoRI, BamHI) are created by chance."

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6. Addressing Specific User Questions

6.1 Adapter Bias in RBNS vs. Plasmid Libraries

User Query: "I would be curious to see if the adapter sequence could interfere and prevent observation of specific sequences."

Expert Insight: In the Dominguez et al. paper, adapter bias was a major concern. They computationally masked reads where the random region paired with the fixed adapter. However, physical interference (blocking binding) remains a blind spot.

In your **Plasmid RBNS (Aim 2)**, the adapter bias profile will be *different* but still present. The RNA will be folded *in vivo* during transcription.

- **Mitigation Strategy:** We will utilize the **RNAfold** algorithm (ViennaRNA package) in the pipeline to predict secondary structures of the full construct (Adapter + Insert + Adapter). Any design where the "Bait" motif is predicted to pair with the Adapter with $\Delta G < -5 \text{ kcal/mol}$ will be discarded from the library *in silico* before synthesis. This "Negative Design" ensures the motif remains accessible.

6.2 Full-Length Proteins vs. RBDs

User Query: "Given they were considering bipartite motifs, could there be missing interactions from the full length proteins?"

Expert Insight: Absolutely. Dominguez et al. primarily used isolated RBDs (e.g., just the RRM domains) to ensure solubility and ease of purification. However, full-length RBPs often contain disordered "Low Complexity Regions" (LCRs) that drive phase separation (e.g., FUS, TDP-43) or inter-domain linkers that facilitate bipartite binding spanning 20+ nucleotides.

- **Impact on Study:** eCLIP uses the full-length endogenous protein. If a "Discrepant Peak" is actually a long-range bipartite interaction requiring the full protein, RBNS (using RBDs) would miss it.
- **Control:** In Aim 1 (Computational), we will scan "Discrepant Peaks" for **spaced k-mers** (e.g., UGU...N(10-20)...UGU). If we find enrichment of these spaced motifs, it suggests the discrepancy is due to *protein architecture* (domain spacing), not modifications. We must rule this out before claiming a modification effect.

6.3 GC Content and Stem Preferences

User Query: "Could the stem/paired-unpaired preferences... be directly explained by the GC content?"

Expert Insight: High GC content naturally promotes stable secondary structures (3 H-bonds vs 2 for AU). A naive analysis might mistake a "GC-preference" for a "Structure-preference."

- **Correction:** The Dominguez analysis used *Z-scores* which normalize for background composition, and they explicitly modeled structure probability (P_{paired}) independent of sequence. However, modifications like m5C *increase* the stability of GC-pairs even further. Our pipeline will calculate the "Modified Free Energy" (using parameters for m6A/m5C in RNAsoft or similar tools) to see if the modification pushes the GC-rich stem over a stability threshold that creates a binding site (or destroys one).

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7. Conclusion

This research plan proposes a rigorous, bi-modal approach to solving the RBP specificity paradox. By treating the Epitranscriptome as a hidden variable in the thermodynamic equation, we move beyond the "ACGU" view of RNA biology. The combination of data mining (Aim 1) to identify candidates and a cleverly engineered "Endogenous RBNS" assay (Aim 2) to validate them offers a robust path to discovery. The integration of AlphaFold3 provides the atomistic "why" to complement the "what" and "where" of the sequencing data.

This report frames the logic; the next step is to activate Claude Code with the prompt structures defined in Section 5.

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